

Please check the examination details below before entering your candidate information

Candidate surname					Other names				
Centre Number					Candidate Number				

Pearson Edexcel Level 3 GCE

Mock Paper Set 3

Paper reference	8MA0/22
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Mathematics

Advanced Subsidiary

PAPER 22: Mechanics

You must have: Mathematical Formulae and Statistical Tables (Green), calculator	Total Marks
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Candidates may use any calculator allowed by Pearson regulations. Calculators must not have the facility for symbolic algebra manipulation, differentiation and integration, or have retrievable mathematical formulae stored in them.

Instructions

- Use **black** ink or ball-point pen.
- If pencil is used for diagrams/sketches/graphs it must be dark (HB or B).
- **Fill in the boxes** at the top of this page with your name, centre number and candidate number.
- Answer **all** questions and ensure that your answers to parts of questions are clearly labelled.
- Answer the questions in the spaces provided
– *there may be more space than you need.*
- You should show sufficient working to make your methods clear. Answers without working may not gain full credit.
- Unless otherwise indicated, wherever a value of g is required, take $g = 9.8 \text{ m s}^{-2}$ and give your answer to either 2 significant figures or 3 significant figures.

Information

- A booklet 'Mathematical Formulae and Statistical Tables' is provided.
- The total mark for this part of the examination is 30. There are 5 questions.
- The marks for **each** question are shown in brackets
– *use this as a guide as to how much time to spend on each question.*

Advice

- Read each question carefully before you start to answer it.
- Try to answer every question.
- Check your answers if you have time at the end.

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1.

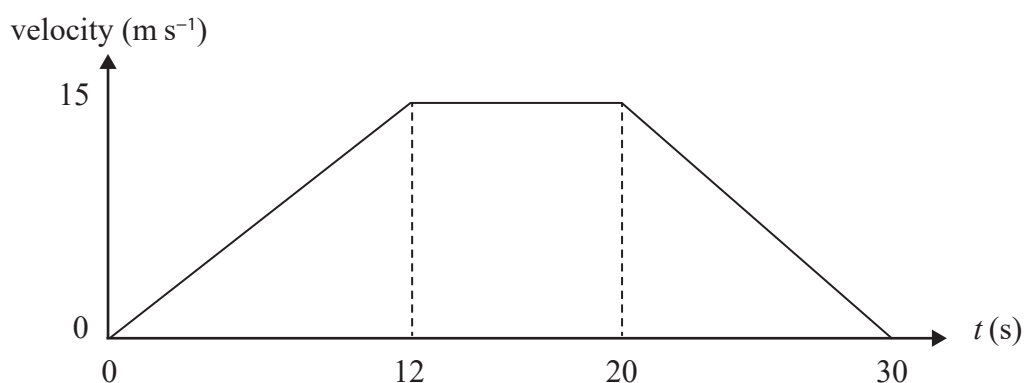


Figure 1

Figure 1 shows a velocity-time graph for the motion of a car.

The car is moving along a straight horizontal road between two points A and B ,

- at time $t = 0$, the car starts from rest at A
- the car accelerates at a constant rate for 12 seconds until it reaches a velocity of 15 ms^{-1}
- the car then maintains a constant velocity of 15 ms^{-1} for T seconds
- the car then decelerates at a constant rate
- the car comes to rest at B

(a) Write down the value of T . (1)

(b) Find the acceleration of the car between $t = 0$ and $t = 12$ (2)

(c) Find the distance AB . (3)



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Question 1 continued

Lined area for writing the answer to Question 1.

(Total for Question 1 is 6 marks)



2.

[In this question, $\mathbf{i}$ and $\mathbf{j}$ are horizontal unit vectors.]

A particle P is on a smooth horizontal plane.

The particle P is at rest under the action of three forces:

$$\mathbf{F}_1 = (-\mathbf{i} + 2\mathbf{j}) \text{ N},$$

$$\mathbf{F}_2 = (3\mathbf{i} - 4\mathbf{j}) \text{ N},$$

$$\mathbf{F}_3 = (c\mathbf{i} + 2\mathbf{j}) \text{ N}, \text{ where } c \text{ is a constant.}$$

Given that P is in equilibrium,

(a) find the value of c .

(2)

The force $\mathbf{F}_3$ is now removed.

Given that the mass of P is 0.5 kg,

(b) find the acceleration of P , giving your answer in terms of $\mathbf{i}$ and $\mathbf{j}$.

(2)



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Question 2 continued

Lined area for writing the answer to Question 2.

(Total for Question 2 is 4 marks)



3.

In this question you must show all stages of your working.**Solutions relying on calculator technology are not acceptable.**

A particle P moves along the x -axis.

At time t seconds, $t \geq 0$

- the distance of P from the origin O is x metres
- the velocity of P is $v \text{ ms}^{-1}$

When $t = 0$, P passes through O .

Given that

$$v = 6t^2 + 4t + 1$$

- (a) find the acceleration of P in terms of t , where $t \geq 0$

(2)

- (b) find the distance of P from O when $t = 3$

(3)



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Question 3 continued

Lined area for writing the answer to Question 3.

(Total for Question 3 is 5 marks)



- Using the model,

- (b) find the total time from when the stone leaves A to when the stone first hits the ground.
- (3)**

Two possible refinements of the model are

- to include air resistance
 - to use a more accurate value for g
- (c) State **one other** possible refinement of the model that would make it more realistic. **(1)**

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Question 4 continued

Lined area for writing the answer to Question 4.

(Total for Question 4 is 6 marks)



5.

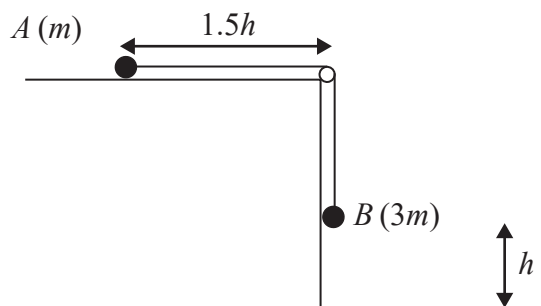


Figure 2

Two particles, A and B , are connected by a string.

Particle A has mass m .

Particle B has mass $3m$.

The string passes over a small smooth light pulley.

The pulley is fixed at the edge of a smooth horizontal table.

Particle A is held at rest on the table at a distance of $1.5h$ from the pulley.

Particle B hangs freely with the string taut, at a height h above the horizontal floor, as shown in Figure 2.

Particle A is released from rest and begins to move along the table.

In an initial model,

- the string is light and inextensible
- air resistance is ignored
- B hits the floor with speed V

Using the model,

(a) show that the tension in the string is $\frac{3mg}{4}$ (5)

(b) find V in terms of g and h (2)

One limitation of the model is that it ignores air resistance.

(c) State **one other** limitation of the model. (1)

In a refinement to the model,

- air resistance is **not** ignored
- B hits the floor with speed W

(d) State which is greater, V or W , giving a reason for your answer. (1)



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Question 5 continued

Lined area for writing the answer to Question 5.



Question 5 continued

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(Total for Question 5 is 9 marks)

TOTAL FOR MECHANICS IS 30 MARKS

